

A Mathematical Model of Translational Coordinate-Motion Puzzles

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Abstract

Coordinate-motion is a geometric concept found in mechanical puzzles. For a coordinate-motion puzzle to come apart or disassemble, three or more pieces must move simultaneously. Here we present a mathematical model of coordinate-motion involving piece translations only, piece rotations are not allowed. We derive equations satisfied by the piece movement vectors, and solve these equations for regular polygons as well as Platonic and rhombic solids. We give examples of new puzzles discovered by solving these equations.

Introduction

The *Magic Disk* [3] is an early example of a coordinate-motion puzzle (Figure 1). This puzzle consists of three identical pieces; it can be assembled into a three-layer hexagon by placing the pieces as shown in Figure 1(b) and moving all three together simultaneously. After assembly, no other piece movements are possible and it can seem welded shut if you attempt to move one piece by itself. You can 3D print a copy of this puzzle here [5].

Three pieces moving simultaneously is unusual and unexpected; this novelty tends to make these puzzles tricky to take apart. The simultaneous movement of many pieces does not happen by accident but is an integral part of the design. Stewart Coffin coined the term “coordinate-motion” for the situation where at least three pieces must move simultaneously, and invented dozens of examples [2].

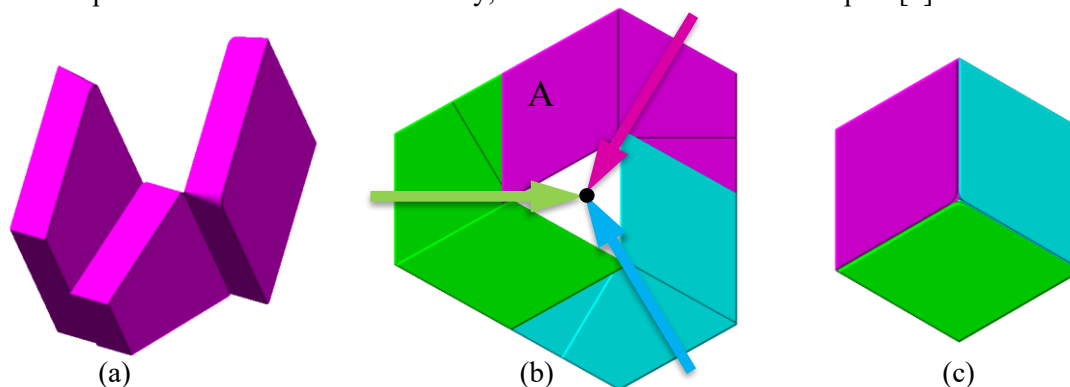


Figure 1: *Magic Disk* puzzle: (a) one piece (b) three pieces apart (c) assembled puzzle. The vectors show the direction of movement for the piece of the same color.

In [1], we provided a definition of coordinate-motion and analyzed many examples. Here we start with that definition and present a complete mathematical model where only piece translations are considered. We will apply this theory first to 2D coordinate-motion puzzles, like that in Figure 1. Next, we will show how to convert a convex polyhedron into a coordinate-motion puzzle. Finally, we will solve for piece movements for the Platonic Solids as well as the rhombic dodecahedron and triacontahedron (we will refer to the latter two as rhombic solids).

When we refer to the *assembly* of a puzzle, we refer not to the assembled state but rather the movement of pieces toward that state. Conversely, the *disassembly* refers to the opposite action—the movement of pieces resulting in the object coming apart.

We define a puzzle as *strongly interlocking* if, in the assembled state, no piece can move when all other pieces are prevented from moving. This is a very restrictive condition; for example, most burr puzzles are not strongly interlocking. Strongly interlocking indicates that the piece movements are severely limited. All puzzles in this paper are strongly interlocking.

Theory of Coordinate-Motion

Consider a puzzle with n pieces. We always use three spatial dimensions, so all vectors are in $\mathbb{R}^3$. Consider the puzzle in the assembled state. Let c_i be some point on piece i , and $\vec{c}_i$ the vector from the origin to c_i . For example, c_i could be the center of mass. In general, we will choose c_i to simplify the movement constraints. We choose our coordinate origin so that the sum of $\vec{c}_i$ over all pieces is $\vec{0}$ (the zero vector). We can fix the puzzle orientation by requiring (for example) that the x -axis pass through c_1 .

The set of piece translations is represented by a set of piece movement vectors (PMVs), $\{\vec{p}_i, i = 1, \dots, n\}$, meaning that piece i moves from c_i to $c_i + \vec{p}_i$. We then define the disassembly specified by $\vec{p}_i$ as *coordinate-motion* if the set $\{\vec{p}_i, i = 1, \dots, n\}$ contains at least three different vectors. The coordinate-motion distinction is only useful for strongly interlocking puzzles.

In what follows, we will reduce the disassembly of a puzzle to the movement of the n points subject to linear movement constraints. In this respect, our theory is encompassed by Rigidity Theory, which considers a system of points and rigid rods connecting them, together with constraints on their movement. The supplement contains further details on the connection with Rigidity Theory.

However, there is one part of our theory where the 3D shape of the pieces remains important: we cannot allow piece movements which result in pieces overlapping or intersecting one another. Movement constraints which prevent this we call *overlap constraints*. In general, an overlap constraint will depend on the 3D shape of the pieces.

Consider now the *Magic Disk* puzzle. We first simplify the puzzle into a “rods and holes” version which has the same translational movement constraints. In Figure 1, we take the part of the magenta piece labeled “A”, and form it into a rod, which fits into a hole in the green piece. We end up with the puzzle in Figure 2 with three identical pieces.

The idea of simplifying a puzzle into a “rods and holes” model is due to Stewart Coffin [2]. The value of the rods and holes model is that it can be generalized, as we shall see. Although we consider the puzzle in Figure 2, the constraints we derive are general and will be applied to all puzzles in the remainder of this paper.

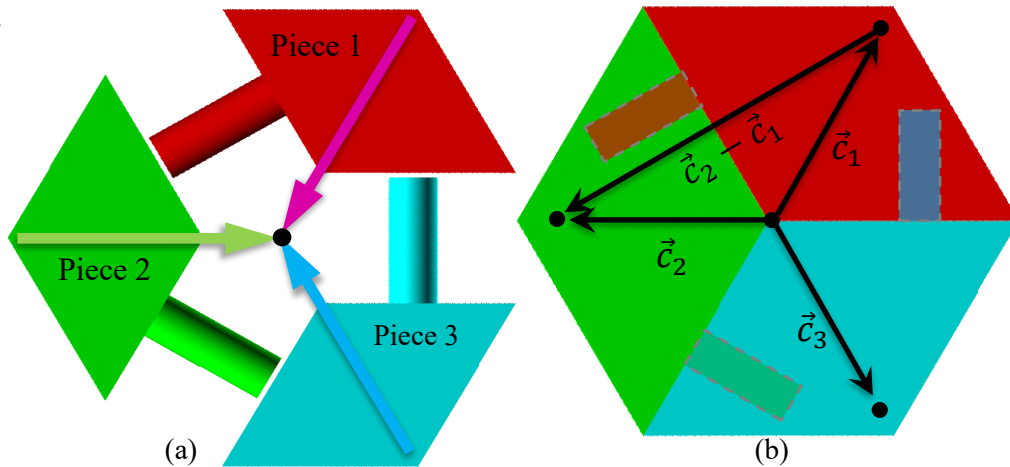


Figure 2: The Magic Disk as a “rods and holes” puzzle (a) unassembled (b) assembled.

Let $\vec{n}_{ij}$ be a vector parallel to the rod joining pieces i and j . The relative movement of pieces i and j , $\vec{p}_j - \vec{p}_i$, must be parallel to $\vec{n}_{ij}$. If we choose $\vec{c}_i$ such that $(\vec{c}_j - \vec{c}_i) \parallel \vec{n}_{ij}$, we can write this constraint as a cross product,

$$(\vec{p}_j - \vec{p}_i) \times (\vec{c}_j - \vec{c}_i) = \vec{0}. \quad (1)$$

In our applications, there is a clear choice of $\vec{c}_i$ which satisfies $(\vec{c}_j - \vec{c}_i) \parallel \vec{n}_{ij}$. But in general, there may not exist a $\vec{c}_i$ which satisfies $(\vec{c}_j - \vec{c}_i) \parallel \vec{n}_{ij}$ for all i and j . This is discussed in more detail in the supplement.

Dynamically, we fix the origin at the centroid of all c_i by the requirement

$$\sum (\vec{p}_i + \vec{c}_i) = \sum \vec{p}_i = \vec{0}. \quad (2)$$

Finally, the pieces must not overlap. While equation (1) specifies that $\vec{p}_j - \vec{p}_i$ and $\vec{c}_j - \vec{c}_i$ are collinear, in addition this relative movement can proceed in only one direction. This constraint may be written as

$$(\vec{p}_j - \vec{p}_i) \cdot (\vec{c}_j - \vec{c}_i) \geq 0. \quad (3)$$

Constraint (3) guarantees that pieces i and j do not overlap, or that their intersection is empty. When we limit piece movements to translations, the overlap constraint is captured by this simple inequality.

If we include piece rotations, the overlap constraint (3) will become much more complicated. For then to guarantee no overlap we must calculate the intersection between pieces i and j after each has undergone a different 3D rotation. For 3D pieces and arbitrary 3D rotations, this is a difficult task.

To summarize, we define an abstract coordinate-motion puzzle by:

1. A set V of n pieces identified by a set of distinct vectors $\{\vec{c}_i, i = 1, \dots, n\}$.
2. A set E of e unordered pairs of interacting pieces. An edge $\{i, j\} \in E$ if and only if the relative motion of pieces i and j is constrained to be parallel to $\vec{c}_j - \vec{c}_i$.
3. The interaction graph $G = (V, E)$ must be connected.

After choosing V, E and G the puzzle is defined and fixed. The piece movements vectors $\vec{p}_i$ satisfy linear equalities (1) and (2), and the linear inequality (3). To find all PMV's, we must solve (1)-(3) for $\vec{p}_i$.

Note that $\vec{p}_i = \vec{0}$ is a trivial solution to constraints (1)-(3), this represents the puzzle in the assembled state. One can also check that $\vec{p}_i = \vec{c}_i$ is a solution, and in fact any non-negative multiple of $\vec{c}_i$ is a solution. The solution $\vec{p}_i = t\vec{c}_i$ for any $t \geq 0$ is known as *uniform expansion*, because it involves every piece moving radially outward. We use t here because we think of it as time; when $t = 0$ the puzzle is assembled and as t increases the puzzle comes apart or disassembles.

Equations (1) and (2) are linear, homogeneous equations for the unknowns $\vec{p}_i$. The constraints (3) are linear homogeneous inequalities. Because of the inequalities the solution is not a vector space, but an unbounded subset called a *convex cone*. Solving these equations involves a branch of linear algebra involving inequalities and convex sets [4]. We present a summary of the properties of solutions to constraints (1)-(3). Further details are presented in the supplement.

A convex cone Ψ is defined by the property that if $x, y \in \Psi$ and $t \geq 0$, then $x + y \in \Psi$ and $tx \in \Psi$ (the set Ψ is closed under addition and scalar multiplication, but only for non-negative scalars). An *extreme ray* of a convex cone Ψ is an element $r \in \Psi$ such that $r = x + y$ with $x, y \in \Psi$ implies $r = tx$ or $r = ty$ for some $t \geq 0$. In the supplement, we prove that the solution space is a *pointed* convex cone.

We work in 3D, so in $\vec{p}_i$ we have $3n$ unknowns and in constraints (1), (2) and (3) we have $3e$, 3 and e conditions, respectively. The dimension of the solution space (denoted by d) is the rank of the null space of the linear homogeneous equations (the supplement explains why the linear inequalities may be ignored in calculating d). Because $\vec{p}_i = t\vec{c}_i$ is a solution, the dimension d is at least 1.

In the supplement, we show that the solution to equations (1)-(3) can be written as a sum of extreme rays $\vec{r}_{ij}$,

$$\vec{p}_i = \sum_{j=1}^D t_j \vec{r}_{ij} \text{ for } t_j \geq 0 \quad (4)$$

(where in general $D > d$). Equation (4) appeared in [1, Equation (1)]; here we provide this general solution with a firm mathematical foundation in terms of extreme rays. The supplement contains details on this.

If constraint (3) is satisfied as an equality, it must be that $\vec{p}_i = \vec{p}_j$, pieces i and j are moving together as one unit. In the supplement, we show that an extreme ray must satisfy at least $d - 1$ constraints (3) as an equality. Thus, we uncover the interesting fact that when $d > 1$, extreme rays must involve multiple pieces moving together as a unit. At this point we move on to 2D and 3D examples.

Polygon Models of Coordinate-Motion

We now generalize the rods and holes puzzle in Figure 2 to a regular polygon. We first convert the polygon into a 3D object by giving it a finite thickness. We then dissect it into n pieces by cutting it into sectors of angle $2\pi/n$ and insert a rod or hole perpendicular to each interior face (Figure 3(a)). Although the pieces can only move in the xy plane, we retain the z dimension in order to apply the cross-product formulation in Equation (1). We also identify $\vec{p}_{n+1}$ with $\vec{p}_1$. Finally, note that in the $n = 3$ and $n = 4$ cases we have extended each piece outward to make it larger without changing the mechanism. For example for $n = 3$ the assembled shape is a hexagon (Figure 2) rather than an equilateral triangle.

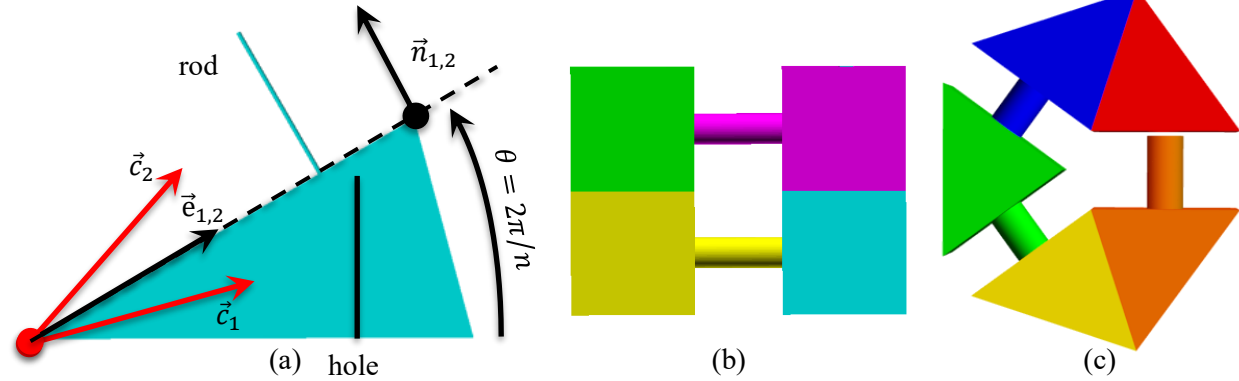


Figure 3: (a) One piece in the polygon model. (b) An extreme ray for the four piece puzzle ($n=4$). (c) An extreme ray for the five piece puzzle ($n=5$).

Let $\vec{c}_i$ be the radial vector as shown in Figure 3(a).

$$\vec{c}_i = \left(\cos \frac{\pi(2i-1)}{n}, \sin \frac{\pi(2i-1)}{n}, 0 \right), i = 1, 2, \dots, n$$

The interaction graph G is the n -cycle graph. Using trigonometric identities, we calculate that

$$\vec{c}_{i+1} - \vec{c}_i = k \vec{n}_{i,i+1} \text{ where } \vec{n}_{i,i+1} = \left(-\sin \frac{2\pi i}{n}, \cos \frac{2\pi i}{n}, 0 \right)$$

and $k = \sin(\pi/n)$ is a constant. In addition, only the z -term in the cross product is non-zero, so we can rewrite Equations (1)-(3) in a simpler form, in agreement with [1, supplement].

$$(\vec{p}_{i+1} - \vec{p}_i) \cdot \vec{e}_{i,i+1} = 0 \text{ where } \vec{e}_{i,i+1} = \left(\cos \frac{2\pi i}{n}, \sin \frac{2\pi i}{n}, 0 \right) \quad (5)$$

$$\sum \vec{p}_i = \vec{0} \quad (6)$$

$$(\vec{p}_{i+1} - \vec{p}_i) \cdot \vec{n}_{i,i+1} \geq 0 \quad (7)$$

Equations (5), (6) constitute a set of $n+2$ linear homogeneous equations for $2n$ unknowns $\vec{p}_i$. The resulting linear system has full rank, and the solution space has dimension $d = n - 2$. We now present the solutions for the cases $n = 3, 4$, and 5 . For the $n = 3$ case (Figure 2) we have only uniform expansion.

The four piece puzzle ($n = 4$) comes apart in halves in two ways, one is shown in Figure 3(b). This movement is not coordinate-motion since it involves only two groups of pieces moving in opposite

directions. Two pairs of pieces move as a unit. The piece movements in Figure 3(b) form one extreme ray, the other extreme has the two upper (lower) pieces moving up (down) together.

The $n = 5$ puzzle can come apart with two pairs of pieces moving together, as shown in Figure 3(c). This is also an extreme ray, and the general solution can be written as the sum of this solution together with rotations of this solution by multiples of $2\pi/5$. We note that $d = 3$ and $D = 5$; five extreme rays are needed to represent all possible movements. In general, there is no guarantee that extreme rays are linearly independent.

A summary of the properties of these three puzzles is given in Table 1. “Piece groupings” indicates how the pieces are grouped in the extreme rays. “Halves” means that the puzzle has this many ways to come apart in two parts. We can think of the n piece polygon constraints as a subset of the constraints for the n polyhedron pieces which surround a vertex of degree n . For example, we anticipate that a polyhedron with all vertices of degree 3 will support only uniform expansion ($d = 1$).

Table 1: List of polygons converted to coordinate-motion puzzles. d is the dimension of the convex cone. “Halves” indicates the number of solutions which come apart in two parts.

Polygon	Pieces (n)	Edges	Vertex degree	d	Piece groupings	Halves	Associated puzzles
Equilateral triangle	3	3	3	1	uniform expansion, $\{1,1,1\}$	0	<i>Magic Disk</i>
Square	4	4	4	2	$\{2,2\}$	2	
Regular pentagon	5	5	5	3	$\{2,2,1\}$	0	

Polyhedron Models of Coordinate-Motion

We will now show how to convert any (convex) polyhedron into a coordinate-motion puzzle. We begin with a polyhedron with n faces and e edges. We consider the Platonic solids as well as the rhombic dodecahedron and rhombic triacontahedron (the rhombic solids).

We first dissect the polyhedron into n identical pieces, one for each face, and then add rods and holes to the interior faces. This is the *rods and holes* procedure for a polyhedron. For every edge $\vec{e}_{ij}$ between faces i and j , consider the triangle T_{ij} defined by $\vec{e}_{ij}$ and the center. Let $\vec{n}_{ij}$ be a normal vector for this triangle. We slice the polyhedron into n pieces by cutting it along every triangle T_{ij} and add rods and holes perpendicular to each interior face.

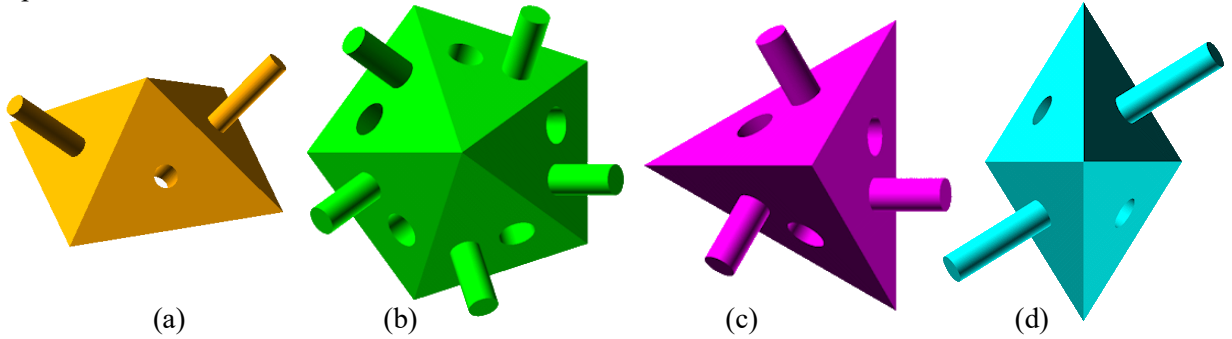


Figure 4: Example pieces of polyhedra converted into coordinate-motion puzzles for (a) cube, (b) dodecahedron, (c) icosahedron and (e) rhombic triacontahedron.

In most cases, we need to add both a hole and rod to each interior face. Figure 4 shows the resulting puzzle pieces for five polyhedra. When e is evenly divisible by n , we need only add either a hole or a rod to each interior face. The rods and holes guarantee that pieces i and j can only move directly away from each other, parallel to $\vec{n}_{ij}$. Here, the interaction graph G is the dual polyhedron graph.

Let $\vec{c}_i$ be the vector from the center of the polyhedron to the center of face i . The key property required of constraints (1) and (3) is

$$(\vec{c}_j - \vec{c}_i) \parallel \vec{n}_{ij}. \quad (8)$$

The Platonic and rhombic solids are edge- and face-transitive, which implies that equation (8) is satisfied.

We have now converted each Platonic and rhombic solid into an n -piece coordinate-motion puzzle, constraints (1)-(3) must be solved to discover the PMVs. If the polyhedron has n faces and e edges we have $3n$ unknowns in the vectors $\vec{p}_i$, and $3e + 3$ equations in (1) and (2). For all polyhedra, $3e + 3 > 3n$, so equations (1) and (2) appear to be overdetermined. Yet we know there is at least one solution, uniform expansion, so evidently the set of linear equations (1) and (2) are linearly dependent.

We have written a solver in Mathematica for constraints (1)-(3). Our solver can be downloaded here [6]. Given a polyhedron, our solver can calculate d and find extreme rays. Table 2 shows the results of the solver for the Platonic and rhombic solids.

For the tetrahedron, cube and dodecahedron, all vertices have degree 3, and only uniform expansion is possible. The octahedron comes apart in halves three ways, separating along coordinate axes when the six vertices are placed on the coordinate axes.

The first interesting case is the rhombic dodecahedron, which comes apart in four groups of three pieces. The solution space of the icosahedron has dimension 9 and can come apart in a large variety of ways. In Table 2, “piece groupings” indicate the sets of pieces that move together in the solution (an extreme ray). For example, $\{3,3,3,3\}$ indicates that four triplets of pieces move together, and uniform expansion is always $\{1,1,1, \dots, 1\}$.

Table 2: List of Platonic and rhombic solids converted to coordinate-motion puzzles.

Polyhedron	Pieces (n)	Edges (e)	Vertex degrees	d	Piece groupings	Halves	Associated puzzles
Tetrahedron	4	6	3	1	only uniform expansion	0	<i>Blast-Hedron 4</i>
Cube	6	12	3	1	only uniform expansion	0	<i>Blast-Hedron 6</i>
Octahedron	8	12	4	3	$\{4,4\}$	3	
Dodecahedron	12	30	3	1	only uniform expansion	0	<i>Blast-Hedron 12</i>
Rhombic dodecahedron	12	24	3 and 4	2	$\{3,3,3,3\}$	0	<i>Splinter 12</i>
Icosahedron	20	30	5	9	see Table 3	0	<i>Dodecahedron Assembly Kit</i>
Rhombic triacontahedron	30	60	3 and 5	4	$\{3,3,3,3,3,1\}$	0	<i>Splinter 30</i>

Associated Puzzles

The rods and holes puzzles make nice examples, and we can also 3D print the pieces in Figure 4 to make physical puzzles. However, the friction between pieces tends to be high and the pieces are also difficult to grasp, making manipulation difficult.

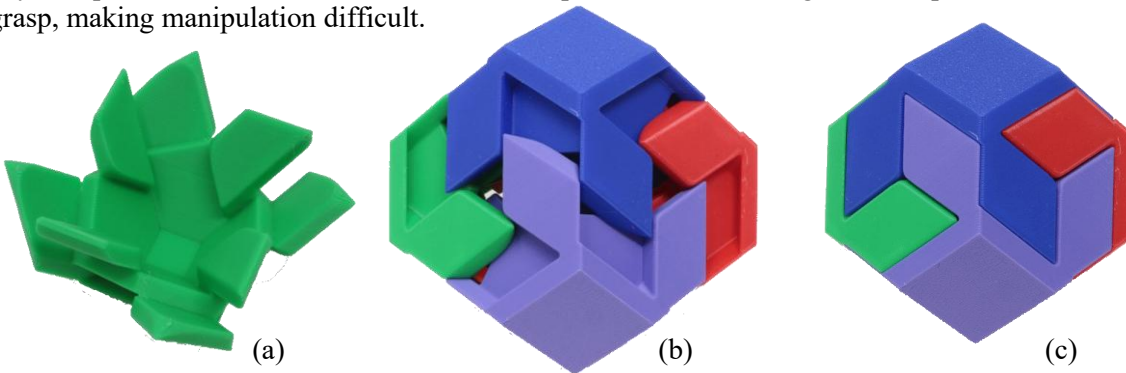


Figure 5: Blast-Hedron 6 puzzle: six identical pieces (a) one piece, (b) assembling (c) assembled.

Previously we found that the three piece, rods and holes example in Figure 2 had the same PMVs as the *Magic Disk*. We can make a similar connection between a rods and holes polyhedron puzzle and the “associated puzzles” in Table 2. All the associated puzzles in Table 2 are new puzzles, discovered by one of us (Nan Ma) in the past two years. These puzzles may be available for purchase here [7].

To create an associated puzzle, we usually begin with the dual polyhedron, because a piece then is associated with a vertex rather than a face. Consider, for example, *Blast-Hedron 6*, which is based on cube PMVs. We begin with the dual solid, an octahedron, with one puzzle piece at each vertex. We then add four wings to each piece so that they interlock across an octahedron face using the *Magic Disk* mechanism. As a final (cosmetic) step, the octahedron is truncated. Figure 5 shows the resulting puzzle [7]. The *Magic Disk* mechanism is also used on each face for *Blast-Hedron 4* and *Blast-Hedron 12*.

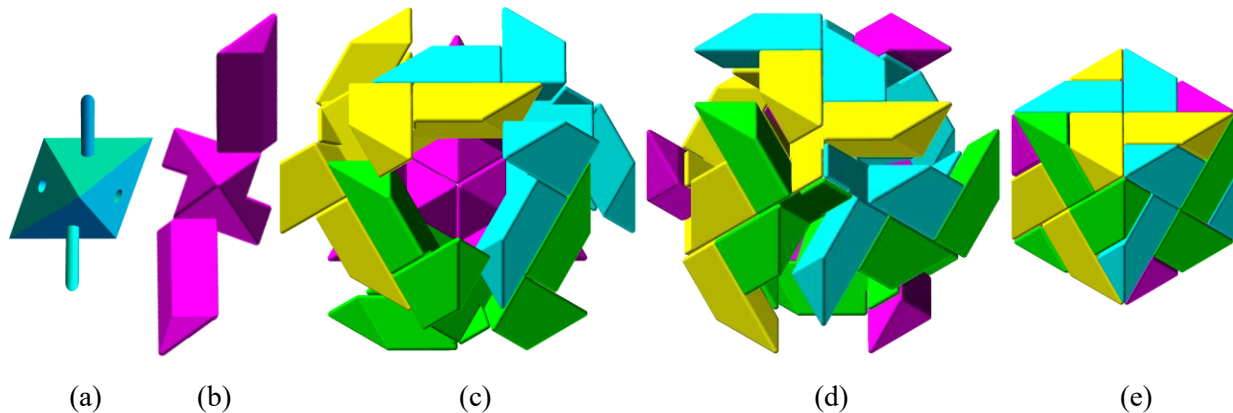


Figure 6: *Splinter 12* puzzle: twelve identical pieces (a) rods and holes piece (b) one piece, (c) disassembly via one extreme ray (d) disassembly via the other extreme ray (e) assembled.

Splinter 12 is based on the PMVs for a rhombic dodecahedron. We begin with its dual, the cuboctahedron, and place each piece at the 12 cuboctahedron vertices. Starting from Figure 6(a), we move the rods away from the center and lengthen them so that three meet to form a cube corner. Each piece extends from one cube vertex to the diagonally opposite vertex. This puzzle comes apart via two PMVs, each involving four groups of three pieces, as shown in Figure 6. The pieces will also move with any non-negative linear combination of these two movements. For example, equal parts of both movements gives uniform expansion. See [8] for a video of this puzzle coming apart, and [9] for a video of *Splinter 30* coming apart.

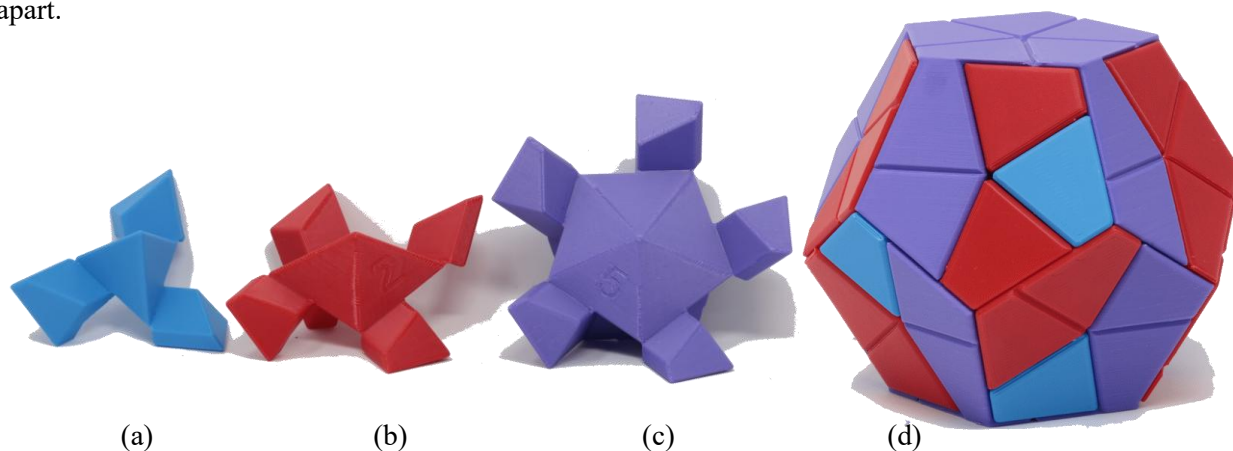


Figure 7: *Dodecahedron Assembly Kit*: (a) single piece (compare with Figure 4(c)), (b) double piece, (c) quintuple piece (d) assembled with 2 singles, 4 doubles and 2 quintuples (Problem #2 in Table 3).

Dodecahedron Assembly Kit is based on the icosahedron, and we begin with its dual, the dodecahedron. A single piece occupies one dodecahedron vertex, a double piece is two connected singles

and a quintuple piece is five singles connected in a ring (Figure 7). Six different piece combinations can assemble into a dodecahedron summarized in Table 3. These puzzles consist of between 8 and 11 pieces.

Dodecahedron Assembly Kit consists of 3 quintuple, 10 double and 6 single pieces; this allows for assembly of all problems in Table 3. This is a very unusual coordinate-motion puzzle, because each problem involves a different set of pieces which assemble into a dodecahedron. The PMVs for each solution are different. The supplement contains diagrams of PMVs for three of the problems in Table 3.

Note that there are many piece combinations (such as 2 quintuple and 5 double pieces) where the end state of pieces fitting together as a dodecahedron is unreachable, because the pieces cannot slide together. Even for the piece combinations in Table 3, if the pieces are not in the correct starting locations the assembly cannot be accomplished.

Table 3: Six problems in the *Dodecahedron Assembly Kit*. “Solutions” counts the number of geometrically distinct solutions to each problem.

Problem No.	Quintuple pieces	Double pieces	Single pieces	Total pieces	Solutions
#1	3	0	5	8	1
#2	2	4	2	8	1
#3	2	3	4	9	1
#4	2	2	6	10	1
#5	1	5	5	11	3
#6	0	10	0	10	1

Summary

Borrowing concepts from Rigidity Theory, we have devised a 3D mathematical model of coordinate-motion involving piece translations. We have added overlap constraints, which prevent pieces from intersecting. These overlap constraints have the form of linear inequalities. The solution space is a subset of a vector space called a (pointed) convex cone. Table 2 summarizes solutions to these equations for PMVs in the case of Platonic solids and rhombic polyhedra. From these solutions we have devised new puzzles which are listed in Table 2. These puzzles may be for sale here [7].

Acknowledgements

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